M8127 User's Manual

This document is the User's Manual for M8127, the interface box for the force/torque sensor (loadcell) manufactured by SRi, Sunrise Instruments Co., Ltd. It's strongly recommended that anyone who uses M8127 should read this document before any operation. Please do not hesitate to contact SRi if there is any question.

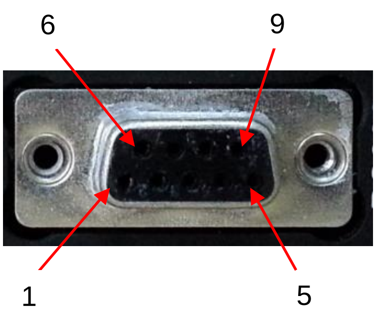
Contents

1. Introduction

The Interface Box M8127 provides bridge excitation, signal conditioning, data acquisition and digital communication to the user's controller or PC via RS232 , CAN Bus or Ethernet. The interface box has 24 analog input channels with programmable gains to allow for low voltage such as strain gage bridge signal. A 24 bit sigma-delta AD converter (16 bit effective) is used to provide high resolution (1/5000 to 1/10000 of full scale) digital signal. The data rate is up to 2KHz. Four 6 axis loadcells can be connected to the Interface Box via four 19 pin LEMO connectors.
M8127 supports two sampling modes: high speed mode(H) and low speed mode(L). In high speed mode, M8127 can sample 18 channels data simultaneously and each channel's sampling rate can up to 2KHZ. In low speed mode, M8127 can sample 24 channels data simultaneously and each channel's sampling rate can up to 1KHZ.

M8127Z is tailored to use with M8130 together. M8130 is the interface box controller, it acquire data from M8127Z and transmit data to user's controller(eg.PC). Up to four M8127Z can be connected to M8130.
M8127Z is most the same as M8127. " M8127 Users' Manual Vx.x-Eng" can be used as reference document of M8127Z, except the follows:

1) The pin definition of DB-9 connector on front panel.



Pin Num#	Definition	Note
1	TDP	Ethernet
2	RX	RS232
3	TX	RS232
4	A	RS485
5	GND	
6	B	RS485
7	TDN	Ethernet
8	RDP	Ethernet
9	RDN	Ethernet

2) CAN Bus is not supported in M8127Z.
3) M8127Z can be connected to M8130 via RS485. Moreover, RS485 is not available for customer.

Specifications:

- Analog
  - # of Channels: 24
  - Programmable gain
  - Automatically adjusting sensor's zero offset
  - Low noise instrumentation amplifiers

- Digital
  - RS232 , CAN Bus and Ethernet
  - 24 bit sigma-delta ADC (16 bit effective), Sampling rate: up to 2 kHz
  - Speed Mode：High speed mode(up to 2 kHz sampling rate) and Low speed mode(up to 1kHz sampling rate)
  - Resolution: 1/5000 to 1/10000 of full scale, when connected to SRI's sensors
  - Programmable system parameters

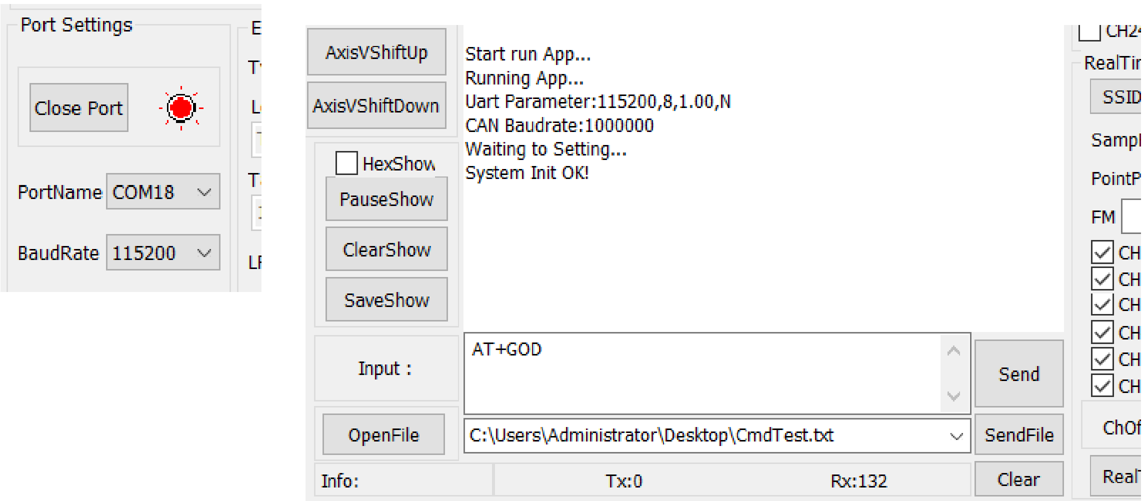
- Frontal Panel
  - Loadcell connector: LEMO FGG.2B.319.CLAD52Z
  - Digital: Standard RS232 connector
  - Power supply: 12 to 36V, 800mA. Power cable - Diameter 3.5mm& Length 2m
  - Indicated lights: Power & Status

2. Quick Start

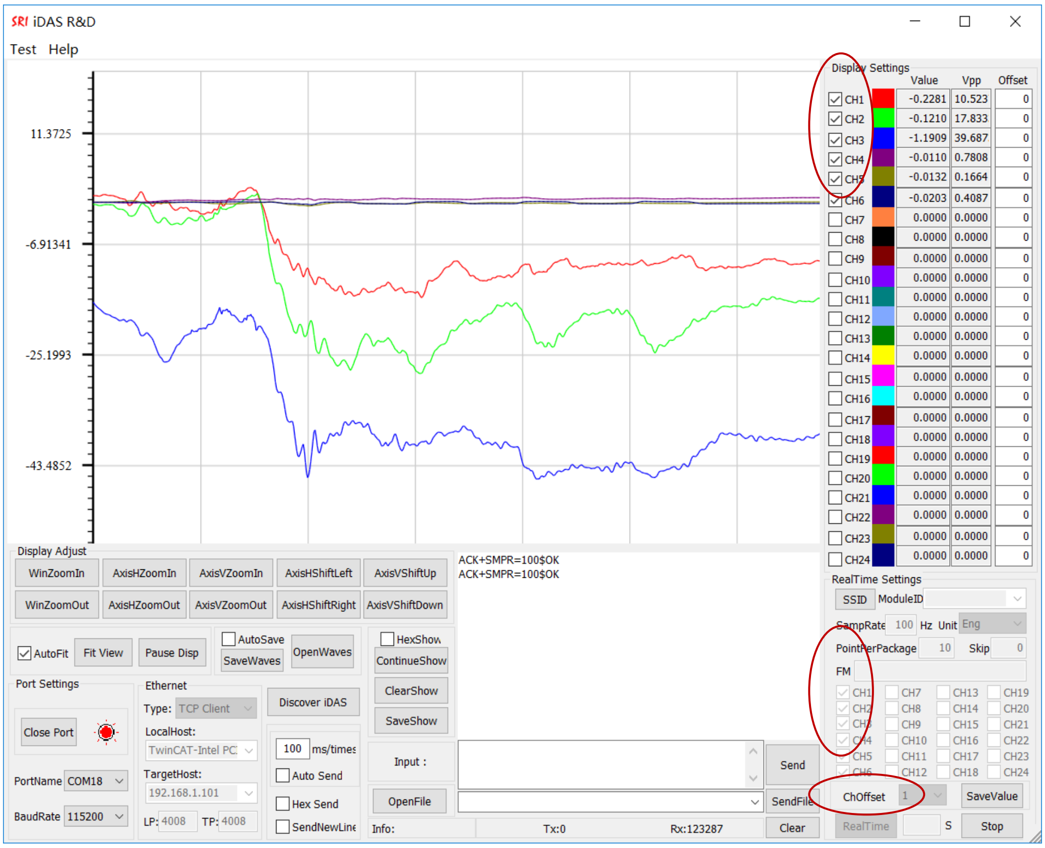
1). Connect the loadcell to M8127 via a LEMO connector.
2). Connect M8127 to PC via RS232/Ethernet/CAN.
3). Provide 24V DC power supply (not included) to the interface box M8127.

2.1 RS232 communication

Step1: Run iDAS R&D.exe. Then select PortName and BaudRate, and click Open Port. If the port is working, the indicated light will turn red.
Step2: Turn power on. Initialization information will appear in the information box.



Step 3: Select Eng in the “Unit” box, check CH1~CH6 boxes to represent FX, FY, FZ, MX, MY, MZ, then click “RealTime” button to start obtaining real-time force curves.

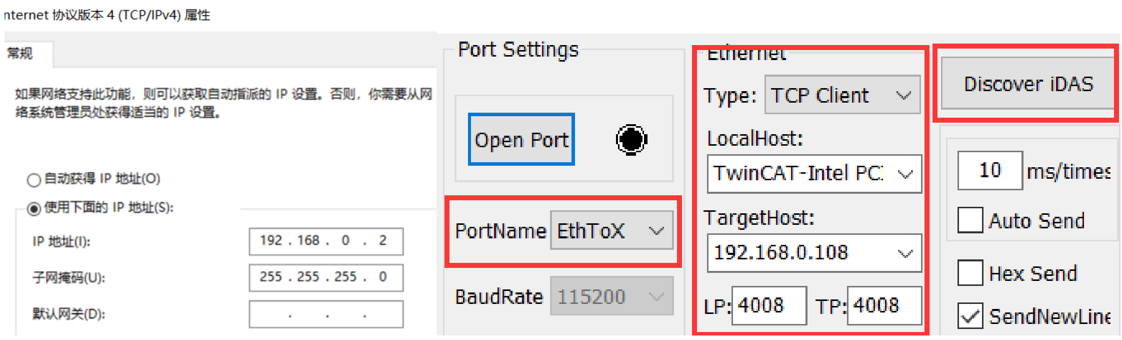


	If the information box doesn’t show initialization information after 1 minute of power-on, please check if: 1) connection and PortName are correct; 2) BaudRate is selected as 115200. If the real time data of CH1~CH6 are incorrect, please check the decoupled matrix and the calculation unit, and refer to Chapters 4.0, 5.3, 5.4 and 7.0 of this Manual.

2.2 Ethernet communication

Step 1: Run iDAS R&D.exe. Then set PortName to EthToX; set Ethernet Type to TCP; select LocalHost (your PC’s Ethernet card).

Click “Discover iDAS” button, software will search M8127. “1 iDAS found” will be shown on screen if M8127 is reached successfully.



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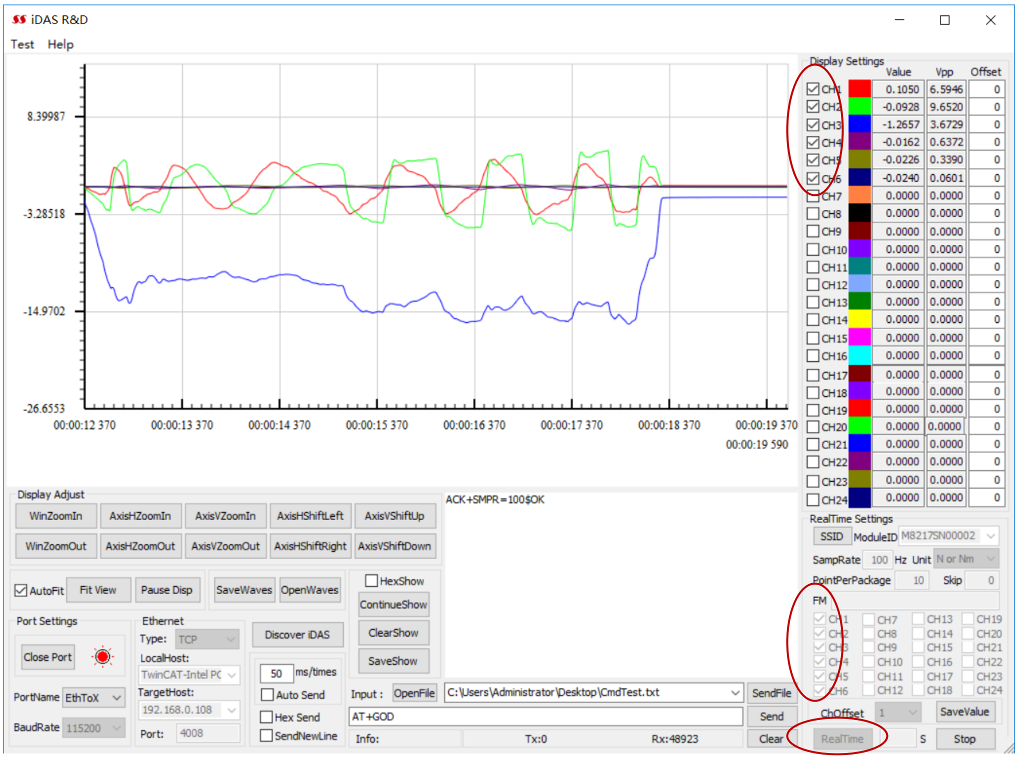
Step 2：Click “Open Port”, the indicated light will turn red.
Step 3：Select Eng in the output Unit box; check CH1~CH6 boxes to represent FX，FY，FZ，MX，MY，MZ; then click “RealTime” button to start obtaining real-time force/torque curves.



If the pop-up window shows “0 iDAS found”, please try 1) click “Discover iDAS” button again； 2) click “ LocalHost”; 3) change the network card and try again.
If the real time data of CH1~CH6 are incorrect, please check the decoupled matrix and the calculation unit, and refer to Chapters 4.0、5.3、5.4 and 7.0 of this Manual.

2.3 CAN Bus communication

Use RS232 communication to configure ID type, ID filter list, and baud rate for CAN BUS (refer to chapters 4, 5.1.6, 5.1.7, and 5.1.8.). Note that CAN BUS cable is not included, the cable can be made based on the pin definition for the DB9 connector in Section 3.2.2. The command should be converted into ASCII. When a command is longer than 8 bytes, the command should be divided into multiple frames.

Example:

RS232 communication

Sent： AT+CIDT=?\r\n

Reply： ACK+CIDT=STD$OK\r\n

Sent： AT+CFIDL=?\r\n

Reply: ACK+CFIDL=NULL$OK\r\n

Sent: AT+CRATE=?\r\n

Reply: ACK+CRATE=BR:1000000$OK\r\n

CAN communication

To convert AT+SMPF=?\r\n to ASCII: 41 54 2B 53 4D 50 46 3D 3F 0D 0A. The first frame is: 41 54 2B 53 4D 50 46 3D, and the second frame is: 3F 0D 0A (or: 3F 0D 0A 00 00 00 00 00)

3. Power Cable, Connectors and LED Lights

3.1 Power cable

M8127 comes with a 2-meters power cable, allowing for DC input of 12～36V, with DC24V recommended. Note that DC power supply is not included. If connected with a SRI six axis loadcell, the consumed power is about 4.5W. The cable color codes are defined as follows:

Color	Definition	Note
Red, Blue, Orange	+24V	+Power, red clip
Black, Brown, Yellow, Green	GND	-Power, black clip
Shield	Shield	The power cable shield is connected to the external case of M8127. To reduce noise, it is recommended to connect the shield to both -Vin (black clip) and the true ground in your test lab.

3.2 Connector definition

3.2.1 19 pin LEMO connector

Pin #	Definition	Note
1	CH1+	Channel 1
2	CH1-	
3	CH2+	Channel 2
4	CH2-	
5	CH3+	Channel 3
6	CH3-	
7	CH4+	Channel 4
8	CH4-	
9	CH5+	Channel 5
10	CH5-	
11	CH6+	Channel 6
12	CH6-	
13~16	N/A	
17	-E	Negative excitation (if support)
18	+E	Positive excitation
19	GND	

3.2.2 Ethernet / RS232 / CAN Bus connector

Ethernet/RS232/CAN bus are included the same DB9 connector.
The pin assignments are as follows:

Pin #	Definition	Note
1	TDP	Ethernet
2	RX	RS232
3	TX	RS232
4	CANH	CAN BUS
5	GND	Ground
6	CANL	CAN BUS
7	TDN	Ethernet
8	RDP	Ethernet
9	RDN	Ethernet

3.3 Indicated lights

There are two indicated lights: PWR (Power) and STA (Status). The conditions of these lights are defined as follows:

PWR	STA	Definition	Instruction
ON		Power is on	
ON	Flickering	System is working properly	
ON	ON	Excitation is abnormal	Check the loadcell cable
OFF	Flickering	System works ok. PWR light may get damaged	Either ignore or repair PWR light
OFF	ON	Excitation is abnormal and PWR light may get damaged	Check the loadcell cable or contact us

4. iDAS R&D Debugging Software

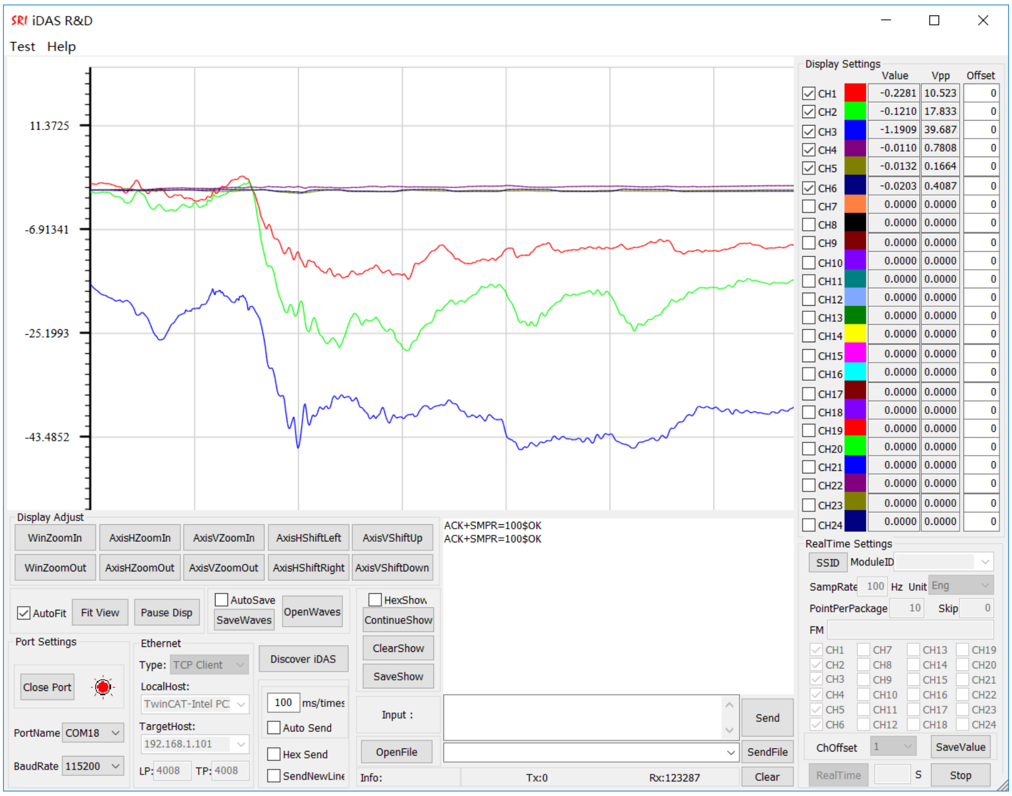
iDAS RD is a debugging software that supports the commands of M8127, which can be used to send a series of commands to M8127 to achieve a special application.

- PC Requirement: WIN 7 and above
- Installation Procedure: Uncompressed iDAS RD
- Support RS232 & Ethernet communication only

4.1 Software Interface

4.3 Realtime data curve

Step 1： Open Port correctly.
Step 2： Select CH1 through CH6 at the lower right corner on screen.
Step 3： Set SampRate to 100Hz， set Unit to N or Nm.
Set PointPerPackage to 10， and put in 0 at Skip.
Step 4： Select CH1 through CH6 at the top right corner on screen.
Step 5： Click “Realtime” to get data from M8127, the real time data will be shown in the window.



Note:

With RS232 communication, SampRate is up to 300 Hz for 6-channel data upload and 1 kHz for 1-channel data upload at BaudRate 115200bps.

With Ethernet communication, SampRate is up to 2K Hz for 6-channel data upload .

If the real-time data shown by iDAS R&D is incorrect, please click Stop and send Commands DCPM and DCPCU to make sure that current matrix coefficients and calculation unit match the sensor calibration report. Refer to 5.3, 5.4 and 6.0 of this Manual.

5. Command

Definitions：

Master： The equipment that send commands to M8127, such as PC or the user's control system. M8127 is called as Slave Equipment.

ASCII Code： America Standard Code for Information Interchange，refer to ISO 646.

M8127 commands are comprised of ASCII codes.

Command structures are shown as follows：

Send to Slave Equipment：

AT+CMD=Parameter\r\n

Response from Slave Equipment：(Except for the command GOD and GSD)

ACK+CMD=Parameter$ResponseCode\r\n

All data that sent to slave equipment must be ASCII code.
All data that received from slave equipment are ASCII code.
Before sent or after received, the data must be converted to or from ASCII

Descriptions：

AT： Frame Header when sending data. All data that are sent to Slave Equipment must be started with AT.

ACK： Frame Header when receiving data. All data that are received from Slave Equipment are started with ACK.

CMD： Command，such as SMPR, etc,.

Parameter： Parameters follow a command.

\r\n： Enter. It denotes the end of Command.

ResponseCode： Response code，such as OK or ERROR.

$：Interval symbols.

Note：

- Parameter '?' denotes that Master is asking for response data from Slave Equipment. Otherwise, Master is sending data to Slave Equipment.
- Response will be sent from Slave Equipment just after the command is executed.

M8127 Command Index

Command	Function	Note
To configure RS232 or CAN Bus or Ethernet		
UARTCFG	To read or set parameters of RS232	
CRATE	To read or set baud rate of CAN Bus	Become available after restart M8127
CIDT	To read or set ID type of CAN Bus	Become available after restart M8127
CFIDL	To read or set ID of CAN Bus	Become available after restart M8127
EIP	Ethernet IP address	Become available after restart M8127
EMAC	Ethernet MAC address	Become available after restart M8127
EGW	Ethernet gateway	Become available after restart M8127
ENM	Ethernet netmask	Become available after restart M8127
System parameters		
SMPRM	To read or set sampling mode (High or Low).	
SMPF	To read or set sampling rate	
To get real-time data from M8127		
SGDM	To set the mode to receive data	
DCKMD	To set the data validation method	
GSD	To get data from M8127 repeatedly	
GOD	To get one package data from M8127	

5.1 Commands to configure RS232/CAN

5.1.1 Parameters of RS232

Description: To read or set parameters of RS232

Command Syntax: AT+**UARTCFG**=Rate,DataBit,StopBit,ParityBit

Command	Possible response(s)
AT+**UARTCFG**=?	Rate,DataBit,StopBit,ParityBit
AT+**UARTCFG**=Rate,DataBit,StopBit,ParityBit	OK/ERROR

Note:
The Master Equipment will receive messy codes after sending a new Baud Rate(X) to Slave Equipment by command UARTCFG. This situation is caused by the different Baud Rate between Master Equipment and Slave Equipment. Therefore, it's recommended that the Baud Rate for the Master Equipment is changed to X and the command UARTCFG is sent to M8127 again to get a correct response.

Parameters

Parameter	Variable Type (Valid Range)	Description
Rate	Unsigned long int ($0\sim2^{32}-1$)	Baud Rate of RS232 in bps. For example 115200bps. Baud Rate of RS232 in M8127 can be 115200bps, 921600bps,460800bps,256000bps,230400bps, 57600bps, 56000bps, 38400bps, 19200bps, 14400bps or 9600bps.
DataBit	int	Number of data bits in RS232 communication. It can be 5,6,7 or 8.
StopBit	float	Number of stop bits in RS232 communication. It can be 0.5,1.0.1.5 or 2.0
ParityBit	char	Parity in RS232 communication. N,O and E denote none, odd and even respectively.

Example:
Send：AT+UARTCFG=?\r\n
Response：ACK+UARTCFG=115200,8,1.00,N$OK\r\n
Send：AT+UARTCFG=115200,8,1.00,N\r\n
Response：Messy code

Master Equipment Baud Rate is changed to the new one:
Send：AT+UARTCFG=115200,8,1.00,N\r\n
Response：ACK+UARTCFG=115200,8,1.00,N$OK\r\n

5.1.2 ID type for CAN Bus

Description: To read or set ID type for CAN Bus
Command Syntax: AT+**CIDT**=Type

Command	Possible response(s)
AT+**CIDT**=?	Type
AT+**CIDT**=Type	OK/ERROR

Note:
The configured ID type will be available after M8127 is restarted.

Parameters

Parameter	Variable Type (Valid Range)	Description
Type	String	The Type can be STD or EXT.STD denotes the standard 11 bits ID and EXT denotes the extended 29 bits ID.

Example:
Send：AT+CIDT=?\r\n
Response：ACK+CIDT=STD$OK \r\n

5.1.3 Baud Rate of CAN Bus

Description: To read or set baud rate of CAN Bus.
Command Syntax: 1. AT+**CRATE**=BR:rate

2. AT+**CRATE**=RP:BS1,BS2,Prescaler

Command	Possible response(s)
AT+**CRATE**=?	1. BR:rate 2. RP:BS1,BS2,Prescaler
1. AT+**CRATE**=BR:rate 2. AT+**CRATE**=RP:BS1,BS2,Prescaler	OK/ERROR

Note:

1. The default Baud Rate of CAN Bus in M8127 is 1Mb/s, and the baud rate can be changed by the command CRATE through two ways.

   1.1 One method is to send “AT+CRATE=BR:rate” to set the Baud Rate, where the rate should be 1Mb/s, 0.8Mb/s, 0.75Mb/s, 0.6Mb/s, 0.5Mb/s, 0.45Mb/s, 0.25Mb/s or 0.125Mb/s.

   1.2 The other method is to send “AT+CRATE=RP:BS1,BS2,Prescaler” to set the Baud Rate. More Baud Rate can be achieved by this method. The Baud Rate is defined as following:

   Baud Rate = 36/((1+ BS1+ BS2)*(1+Prescaler))Mbps

2. Only one method can be used each time.
3. It will be available after M8127 is restarted.

Parameters

Parameter	Variable Type (Valid Range)	Description
BR	String	Keyword
RP	String	Keyword
rate	Unsigned long int ($0\sim2^{32}$-1)	Baud Rate in bps. This parameter can be 1000000, 800000, 750000, 600000, 500000, 450000, 250000 or 125000.
BS1	Unsigned short int (0～65535)	An integer which is through 1 to 16.
BS2	Unsigned short int (0～65535)	An integer which is through 1 to 8.
Prescaler	Unsigned short int (0～65535)	An integer which is through 1 to 1024.

Example:

Send：AT+CRATE=?\r\n
Response：ACK+CRATE=BR:1000000$OK\r\n

Send：AT+CRATE=?\r\n
Response：ACK+CRATE= RP:7,8,20$OK\r\n
Send：AT+CRATE=BR:125000\r\n
Response：ACK+CRATE=BR:125000$OK\r\n

Send：AT+CRATE=RP:7,8,20\r\n
Response：ACK+CRATE=RP:7,8,20$OK\r\n

5.1.4 ID of CAN Bus

Description: To read or set ID of CAN Bus
Command Syntax: AT+**CFIDL**=id1,id2,id3,…,idn

Command			Possible response(s)
AT+**CFIDL**=?			id1,id2,id3,…,idn
AT+**CFIDL**=id1,id2,id3,…,idn			OK/ERROR
Note: One M8127 can have maximum 14 IDs. It will be available after M8127 is restarted.			
Parameters			
Parameter	**Variable Type (Valid Range)**	**Description**	
idn	0～2^{11} or 0～2^{29}	Decimal number	

Example:
Send： AT+CFIDL=128\r\n
Response： ACK+CFIDL=128$OK \r\n
Send： AT+CFIDL=?\r\n
Response： ACK+CFIDL=0,125,126,127,128$OK \r\n

5.1.5 Interval time between frames of CAN Bus

Description: To set (or read) interval time between frames of CAN Bus.
Command Syntax: AT+**CFI**=IntervalTime

Command			Possible response(s)
AT+**CFI**=?			IntervalTime
AT+**CFI**=IntervalTime			OK/ERROR
Note: It will be available after M8127 is restarted.			
Parameters			
Parameter	**Variable Type (Valid Range)**	**Description**	
IntervalTime	0～10000	Interval time in us. The default value in firmware is 0us.	

Example:
Send： AT+CFI=10\r\n
Response： ACK+CFI=10$OK \r\n
Send： AT+CFI=?\r\n
Response： ACK+CFI=10$OK \r\n

5.1.6 Ethernet IP Address

Description: To set Ethernet IP address.
Command Syntax: AT+**EIP**=addr0.addr1.addr2.addr3

Command			Possible response(s)
AT+**EIP**=?			addr0.addr1.addr2.addr3
AT+**EIP**= addr0.addr1.addr2.addr3			OK/ERROR
Note: It will be available after M8127 is restarted.			
Parameters			
Parameter	**Variable Type (Valid Range)**	**Description**	
addr		IP address,eg.192.168.0.108	

Example:
Send： AT+EIP=192.168.0.108\r\n
Response： ACK+EIP=192.168.0.108$OK \r\n
Send： AT+EIP=?\r\n
Response： ACK+EIP=192.168.0.108$OK \r\n

5.1.7 Ethernet MAC

Description: To set Ethernet MAC.

Command Syntax: AT+**EMAC**=addr0-addr1-addr2-addr3-addr4-addr5

Command		Possible response(s)
AT+**EMAC**=?		addr0-addr1-addr2-addr3-addr4-addr5
AT+**EMAC**= addr0-addr1-addr2-addr3-addr4-addr5		OK/ERROR
Note: It will be available after M8127 is restarted.		
Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
addr	String	Ethernet MAC address,eg.12-13-14-15-16-17

Example:
Send: AT+EMAC=12-13-14-15-16-17\r\n
Response: ACK+EMAC=12-13-14-15-16-17$OK \r\n
Send: AT+EMAC=?\r\n
Response: ACK+EMAC=12-13-14-15-16-17$OK \r\n

5.1.8 Ethernet Gateway address

Description: To set Ethernet gateway address.
Command Syntax: AT+**EGW**= addr0.addr1.addr2.addr3

Command		Possible response(s)
AT+**EGW**=?		addr0.addr1.addr2.addr3
AT+**EGW**= addr0.addr1.addr2.addr3		OK/ERROR
Note: It will be available after M8127 is restarted.		
Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
addr	String	Ethernet gateway address,eg.192.168.0.1

Example:
Send: AT+EGW=192.168.0.1\r\n
Response: ACK+EGW=192.168.0.1$OK \r\n
Send: AT+EGW=?\r\n
Response: ACK+EGW=192.168.0.1$OK \r\n

5.1.9 Ethernet netmask

Description: To set Ethernet netmask.
Command Syntax: AT+**ENM**= addr0.addr1.addr2.addr3

Command		Possible response(s)
AT+**ENM**=?		addr0.addr1.addr2.addr3
AT+**ENM**= addr0.addr1.addr2.addr3		OK/ERROR
Note: It will be available after M8127 is restarted.		
Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
addr	String	Ethernet netmask,eg.255.255.255.0

Example:
Send: AT+ENM=255.255.255.0\r\n
Response: ACK+ENM=255.255.255.0$OK \r\n
Send: AT+ENM=?\r\n
Response: ACK+ENM=255.255.255.0$OK \r\n

5.2 System parameters

5.2.1 Sampling Mode

Description: To read or set sampling mode(i.e.high speed mode or low speed mode).
Command Syntax: AT+**SMPRM**=SamplingMode

Command	Possible response(s)
AT+**SMPRM**=?	SamplingMode
AT+**SMPRM**=SamplingMode	OK/ERROR
Note:	

Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
SamplingMode	String	SamplingMode can be H or L. M8127 supports two sampling mode: high speed mode(H) and low speed mode(L). In high speed mode, M8127 can sample 18 channels data simultaneously and each channel's sampling rate can up to 2KHZ. In low speed mode, M8127 can sample 24 channels data simultaneously and each channel's sampling rate can up to 1KHZ.

Example:
Send: AT+SMPRM=?\r\n
Response: ACK+SMPRM=L$OK\r\n

Send: AT+SMPRM=H\r\n
Response: ACK+SMPRM=H$OK\r\n

5.2.2 Sampling Rate

Description: To read or set sampling rate.
Command Syntax: AT+**SMPF**=SampleRate

Command	Possible response(s)
AT+**SMPF**=?	SampleRate
AT+**SMPF**=SampleRate	OK/ERROR
Note:	

Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
SampleRate	Unsigned short int (0～65535)	Sampling rate in Hz. For example, 200.

Example:
Send: AT+SMPF=?\r\n
Response: ACK+SMPF=300$OK\r\n

Send: AT+SMPF=200\r\n
Response: ACK+SMPF=200$OK\r\n

5.2.3 DCPM / Read or set decoupled matrix

Description: To read or set decoupled matrix
Command Syntax:AT+DCPM=Matrix

Command	Possible Response(s)
AT+**DCPM**=?\r\n	Matrix data
AT+**DCPM**=Matrix\r\n	OK/ERROR
Note:	

Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
Matrix	String	The format is as follows.

Example:

Send: AT+DCPM=?\r\n

Response:

ACK+DCPM=(0.000041,-0.020164,-0.000348,0.020287,-0.000145,-0.000047);(-0.000160,-0.011703,-0.000089,-0.011668,-0.000217,0.023526);(-0.031415,-0.000185,-0.032273,0.000010,-0.031708,-0.000481);(-0.000888,-0.000014,0.000951,-0.000006,0.000029,0.000009);(-0.000521,0.000011,-0.000531,-0.000009,0.001061,0.000015);(0.000002,0.000754,-0.000008,0.000753,-0.000007,0.000768)$OK\r\n
Refer to Section **7.0 Loadcell Decoupled Calculation** to obtain the decoupling matrix coefficients and calculation units of sensors

Note:
There are four groups of channels(call them A B C D) in M8127,and 4 pcs of six axis loadcells can be connect to M8127. Different sensor has different decoupling matrix.

Therefore,command DCPM must have different formats:
DCPMA:Group A. DCPMB:Group B. DCPMC:Group C. DCPMD:Group D.

For example, AT+DCPMA=………. is used to configure decoupling matrix of group A.

5.2.4 DCPCU / Calculation unit for decoupled data

Description: To set or read calculation unit.
Command Syntax:AT+DCPCU=Unit

Command	Possible Response(s)
AT+**DCPCU**=?\r\n	Uint
AT+**DCPCU**=Unit\r\n	OK/ERROR
Note:	

Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
Unit	String	MV and MVPV(mV/V)

Example:

Send: AT+DCPCU=?\r\n

Response: ACK+DCPCU=MV$OK\r\n

Send: AT+DCPCU=MVPV\r\n

Response: ACK+DCPCU=MVPV$OK\r\n

Note: Each group must have a same calculation unit.

5.3 Get Real-time Data

5.3.1 Set the mode to receive data

Description: To set the mode to receive data.
Command Syntax: AT+**SGDM**=(CHx,CHx,…,CHx);DataUnit;PNpCH;(FM:p1,p2,p3,…,pn)

Command	Possible response(s)
AT+**SGDM**=(CHx,CHx, …,CHx);DataUnit;PNpCH;(FM:p1,p2,p3,…,pn)	OK/ERROR
Note: The default parameter is (A01,A02,A03,A04,A05,A06);C;1;(WMA:1).	

Parameters

Parameter	Variable Type (Valid Range)	Description
(CHx,CHx,…,CHx)	String	The relevant analog channels. CHx is comprised of three ASCII codes. Note that the parentheses is necessary. For example, if channel 2,5 and 1 are required,(CHx) must be written as (A02,A05,A01), and the uploaded data will be in the order of Channel 2 , Channel 5 and Channel 1.
DataUnit	Character (0～255)	The unit of uploaded data. It's comprised of one character E, V, M or C which denote Engineering unit, mV/V, mV or AD Counts respectively. The method to convert data to Engineering unit value is shown in Section 3.5.
PNpCH	Character (0～255)	Number of data which are desired. PNpCH is comprised of three ASCII codes, and is less than 80. For example, if 20 data points are desired, Num must be written as 20.
(FM:p1,p2,p3,…,pn)		Filter model and relevant parameters. The Weighted Mean Algorithm is supported by M8127. Every sampling point will be averaged with previous N (N<=17) points. FM: Filter model. Set to WMA. p1,p2,p3,…,pn：The weight of each point, where pn is the weight of the latest sampling point. They must be integer. For example,(WMA:1,1,1,2,4) means that the average will be got from five points(D1, D2, D3, D4, D5). The average is defined as: (D5*4+D4*2+D3*1+D2*1+D1*1)/(4+2+1+1+1) Note: Please input (WMA:1) if the Weighted Mean Algorithm is not needed.

5.3.2 To set the data validation method

Description: To set the data validation method when getting one package data from M8127.
Command Syntax: AT+**DCKMD=**Mod

Command	Possible response(s)
AT+**DCKMD=**Mod	OK/ERROR
Note: Software IDAS RD only support	

Parameters

Parameter	Variable Type (Valid Range)	Description
Mod	String	Data validation method: SUM or CRC32. SUM is checksum. CRC32 function in C program is including in the CD-ROM.

5.3.3 To get one package data every time

Description: To get one package data from M8127.

Command Syntax: AT+**GOD**

Command		Possible response(s)
AT+**GOD**		DataFormat
Note: If it's necessary, please use command SGDM to set the mode to receive data.		
Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
DataFormat		Data package, refer to the following for details.

5.3.4 To get data repeatedly

Description: To get data from M8127 repeatedly.
Command Syntax: AT+**GSD**

Command		Possible response(s)
AT+**GSD**		DataFormat
Note: 1. If it's necessary, please use command SGDM to set the mode to receive data. 2. To stop receiving data, send "AT+GSD=STOP\r\n" to M8127.		
Parameters		
Parameter	**Variable Type (Valid Range)**	**Description**
DataFormat		Data package, refer to the following for details.

"DataFormat" is defined as follows：

Frame Header	Package Length	Data Number	Data	CRC32/SUM
0xAA ,0x55	HB,LB	2Byte	(ChNum*N*DNpCH) Byte	4Byte / 1Byte

Note：

- 0xAA ,0x55： Frame header of data package.
  PackageLength：Unsigned short int,16-bits, highest byte first, The length of data of each channel, which equals to 2+ChNum*N*DNpCH+1(SUM check) or 2+ ChNum*N*DNpCH+4(CRC32 check)
  Where
  ChNum: Total number of uploading channels, Defualt value 6
  N: the output unit, Defualt value 4
  DNpCH: Number of sampling points to upload in one package, Defualt value 1
- PackageNo: Every package is labeled, which increases in sequence from 0 to 65535.
- Data： Uploading data with the lowest byte first.
- CRC32/SUM: CRC32 or Checksum. The default data validation method is Checksum. Use Command DCKMD to select the data validation method (Checksum or CRC32).
  CRC32 function (MyCRC_GetCRC32 (uint8_t *pData, uint16_t Length)) in C program is included in the CD-ROM.

  Example:
  Send： AT+DCKMD=SUM\r\n
  Response： ACK+DCKMD=SUM$OK\r\n

Send: AT+GOD\r\n

Response: AA 55 00 63 9B C4 35 4D EB 3F 0A AD 3C 3F 84 6F 14 C0 95 69 6B BD DB 4D 1F BC FA 26 0E 3D 7F D0 9E 3F 91 30 E4 3F 31 BC BA BF 54 8D 81 BC 0A 1D 6F 3B 80 AD B9 3C 61 7B 08 BE 34 1E 63 3F A8 87 2B BF 5A 12 10 BC 06 EC 20 BC DB 64 0A 3D 93 FE 01 40 F3 2E D6 3E 24 6B 92 BF 54 4E F9 3B AB 19 91 BB BD 91 02 3B 53

Where

0xAA ,0x55: Frame header

00 63: PackageLength 2+24*4*1+1=99 bytes;

9B C4: Package No 39108;

Channel 1 Engineering Unit: 35 4D EB 3F, single-precision float (3FEB4D35) converted into 1.838294;

Channel 2 Engineering Unit: 0A AD 3C 3F, single-precision float (3F3CAD0A) converted into 0.737015;

Channel 3 Engineering Unit: 84 6F 14 C0, single-precision float (C0146F84) converted into -2.319306;

Channel 4 Engineering Unit: 95 69 6B BD, single-precision float (BD6B6995) converted into -0.057474;

Channel 5 Engineering Unit: DB 4D 1F BC, single-precision float (BC1F4DDB) converted into -0.009723;

Channel 6 Engineering Unit: FA 26 0E 3D, single-precision float (3D0E26FA) converted into 0.034705

Channel 7 Engineering Unit: 7F D0 9E 3F, single-precision float (3F9ED07F) converted into 1.240738;

Channel 8 Engineering Unit: 91 30 E4 3F, single-precision float (3FE43091) converted into 1.782732;

Channel 9 Engineering Unit: 31 BC BA BF, single-precision float (BFBABC31) converted into -1.458868;

Channel 10 Engineering Unit: 54 8D 81 BC, single-precision float (BC818D54) converted into -0.015814;

Channel 11 Engineering Unit: 0A 1D 6F 3B, single-precision float (3B6F1D0A) converted into 0.003649;

Channel 12 Engineering Unit: 80 AD B9 3C, single-precision float (3CB9AD80) converted into 0.022666;

Channel 13 Engineering Unit: 61 7B 08 BE, single-precision float (BE087B61) converted into -0.133283;

Channel 14 Engineering Unit: 34 1E 63 3F, single-precision float (3F631E34) converted into 0.887180;

Channel 15 Engineering Unit: A8 87 2B BF, single-precision float (BF2B87A8) converted into -0.670039;

Channel 16 Engineering Unit: 5A 12 10 BC, single-precision float (BC10125A) converted into -0.008793;

Channel 17 Engineering Unit: 06 EC 20 BC, single-precision float (BC20EC06) converted into -0.009822;

Channel 18 Engineering Unit: DB 64 0A 3D, single-precision float (3D0A64DB) converted into 0.033788

Channel 19 Engineering Unit: 93 FE 01 40, single-precision float (4001FE93) converted into 2.031163;

Channel 20 Engineering Unit: F3 2E D6 3E, single-precision float (3ED62EF3) converted into 0.418327;

Channel 21 Engineering Unit: 24 6B 92 BF, single-precision float (BF926B24) converted into -1.143895;

Channel 22 Engineering Unit: 54 4E F9 3B, single-precision float (3BF94E54) converted into 0.007608;

Channel 23 Engineering Unit: AB 19 91 BB, single-precision float (BB9119AB) converted into -0.004428;

Channel 24 Engineering Unit: BD 91 02 3B, single-precision float (3B0291BD) converted into 0.001992

SUM Check: 53

6. Get Realtime Data form M8127

Step 1: Use command SMPF to set sampling rate. If the sampling rate is 100Hz:
AT+SMPF=100\r\n

Step 2: Use command GOD to get one package data or command GSD to get data continuously from M8127. Refer to Sections 5.7 and 5.8

Note:

- The parameters set by Command SMPF are saved to M8127, and they are still available after power off.
- If iDAS R&D software is used, M8127 is required to restart (Power off and Power on) before debugging your own codes.
- If the real time data is incorrect, please check the decoupled matrix and the calculation unit, and refer to Section 4.0, 5.3, 5.4 and 7.0 of this Manual.

In default, M8127 will send all of the 24 channels data when get command GOD or GSD. If only some of total channels are desired, please use command SGDM to set.

7. Decoupled Calculation

If the M8127 is purchased together with SRI sensor, the decoupled matrix and calculation unit of SRI sensor have been configured in the M8127. The decoupled matrix and calculation unit can be updated by Command DCPM and DCPCU when necessary.

Decoupled matrix and calculation unit can be found in the calibration report. Two different reports formats will be provided according to the sensor's structure.

7.1 Matrix decoupled loadcell

The decoupled matrix and calculation unit are provided in the calibration report, as shown below:

[DECOUPLED] =

-0.03220	0.49984	0.00136	-1.01398	-0.01208	0.50908
0.00046	0.84855	0.01531	0.02114	-0.03126	-0.86432
1.19167	0.00028	1.20748	0.00224	1.19808	0.00320
-0.06386	-0.00097	0.13028	-0.00009	-0.06523	0.00012
-0.11090	0.00016	-0.00049	0.00075	0.11138	-0.00019
-0.00046	0.08401	-0.00067	0.08304	-0.00089	0.08433

The six axis loads can be decoupled as follows:

Step 1: Obtain the raw data of Channels 1 through 6 into mV
[DAT] = {rawchn1, rawchn2, rawchn3, rawchn4, rawchn5, rawchn6}
where rawchn1, rawchn2, rawchn3, rawchn4, rawchn5 and raw chn6 are in mV

Step 2: To calculate decoupled loads
$[RESULT]^T = [DECOUPLED]*[DAT]^T$
where [RESULT] = {FX,FY,FZ,MX,MY,MZ}. Force Unit: N. Moment Unit: Nm
[DECOUPLED] is the above decoupled matrix

The Commands to input the matrix coefficients and to set the calculation unit are as follows:

AT+DCPMA=(-0.03220,0.49984,0.00136,-1.01398,-0.01208,0.50908);(0.00046,0.84855,0.01531,0.02114,-0.03126,-0.86432);(1.19167,0.00028,1.20748,0.00224,1.198078,0.00320);(-0.06386,-0.00097,0.13028,-0.00009,-0.06523,0.00012);(-0.11090,0.00016,-0.00049,0.00075,0.11138,-0.00019);(-0.00046,0.08401,-0.00067,0.083040,-0.00089,0.08433)
AT+DCPCU=MV

7.2 Structurally decoupled loadcell

The sensitivity provided in the calibration report needs to be converted into a matrix as shown below:

Voltage Calibration

Bridge	Capacity	Zero Offset	Nonlinearity	Hysteresis	Output @ Capacity	Sensitivity	Change
	N/Nm	mV/V	%FS	%FS	mV/V	mV/V/EU	%
FX	-5400	0.0131	-0.08	-0.33	-3.0269	5.6054E-04	0.00
FY	5400	0.0007	0.08	0.27	3.0500	5.6481E-04	0.00
FZ	-10800	0.0001	-0.09	-0.18	-0.7369	6.8230E-05	0.00
MX	-540	-0.0027	-0.09	-0.10	-1.8703	3.4636E-03	0.00
MY	-540	-0.0090	-0.09	-0.09	-1.9014	3.5210E-03	0.00
MZ	432	-0.0099	0.05	0.08	1.9603	4.5378E-03	0.00

Sensitivity unit is mV/V/Eu,. The diagonal elements of the matrix are the inverse of the sensitivities (1/Sensitivity). The calculation unit is mV/V.
The Commands to input the matrix coefficients and to set the calculation unit are as follows:
AT+DCPMA=(1783.9940,0,0,0,0,0);(0,1770.5069,0,0,0,0);(0,0,14656.3095,0,0,0);(0,0,0,288.7169,0,0); (0,0,0,0,284.0102,0); (0,0,0,0,0,220.3711)
AT+DCPCU=MVPV

Four possible conversion formula:

1) Sensitivity unit is mV/V/Eu. The conversion formula is 1/Sensitivity.
Calculation unit is mv/V: AT+DCPCU=MVPV.
2) Sensitivity unit is mV/Eu. The conversion formula is 1/Sensitivity.

1783.9940	0	0	0	0	0
0	1770.5069	0	0	0	0
0	0	14656.3095	0	0	0
0	0	0	288.7169	0	0
0	0	0	0	284.0102	0
0	0	0	0	0	220.3711

Calculation unit is mv: AT+DCPCU=MV.
3) Sensitivity unit is V/V/Eu. The conversion formula is 1/Sensitivity/1000.
Calculation unit mv/V: AT+DCPCU=MVPV.
4) Sensitivity unit is V/Eu. The conversion formula is 1/Sensitivity/1000.
Calculation unit is mv: AT+DCPCU=MV.

7.3 Other Loadcells

Except 6 axis loadcells, other sensors with voltage out put can also be connected to M8127.For calculation in M8127, a matrix is also needed.
Please follow the method described below to get the matrix.

3 Axis loadcell

Voltage Calibration

Bridge	Capacity	Zero Offset	Nonlinearity	Hysteresis	Output @ Capacity	Sensitivity	Change
	N/Nm	mV/V	%FS	%FS	mV/V	mV/V/EU	%
FX	-20000	0.0101	-0.13	-0.28	-2.8941	1.4471E-04	0.00
FY	20000	-0.0027	0.11	0.19	2.8894	1.4447E-04	0.00
FZ	-20000	0.0175	-0.07	-0.27	-0.5441	2.7207E-05	0.00

Sensitivity unit is mV/V/Eu. The diagonal elements of the matrix are the inverse of the sensitivities (1/Sensitivity).

6910.3725	0	0	0	0	0
0	6921.8523	0	0	0	0
0	0	36755.2468	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0

The Commands to input the matrix coefficients and to set the calculation unit are as follows:
AT+DCPMA=(6910.3725,0,0,0,0,0);(0,6921.8523,0,0,0,0);(0,0,36755.2468,0,0,0);(0,0,0,0,0,0);(0,0,0,0,0,0); (0,0,0,0,0,0)
AT+DCPCU=MVPV

Torque Sensor

Voltage Calibration

Bridge	Capacity	Zero Offset	Nonlinearity	Hysteresis	Output @ Capacity	Sensitivity	Change
	Nm	V	%FS	%FS	V	V/EU	%
MZ	100	-0.0049	0.04	0.27	2.0445	2.0445E-02	0.00

Sensitivity unit is V/Eu.
The first row and the first column equals to 1/sensitivity/1000.
Calculation unit is mV.

The Commands to input the matrix coefficients and to set the calculation unit are as follows:
AT+DCPM=(0.048913,0,0,0,0,0);(0,0,0,0,0,0);(0,0,0,0,0,0);(0,0,0,0,0,0);(0,0,0,0,0,0);(0,0,0,0,0,0)
AT+DCPCU=MV

Appendix：